System Setup

System Setup

Programs and Environment

1. Install Python

Visual Studio Code or Cursor

2. Install IDE

3. Install git

4. Config &
Get Course Material

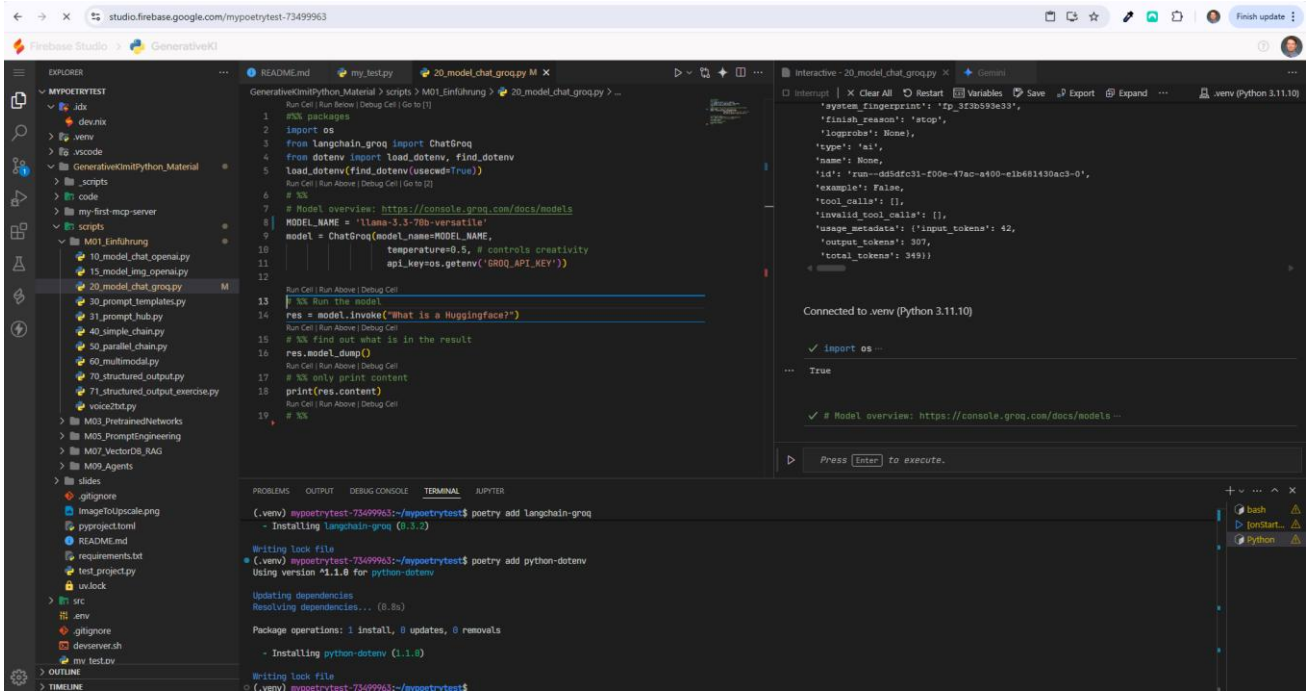
5. Set up Environment

System Setup

Programs and Environment: Cloud

Firebase Studio

Allows coding in the browser



URL: https://studio.firebase.google.com/mypoetrytest-73499963

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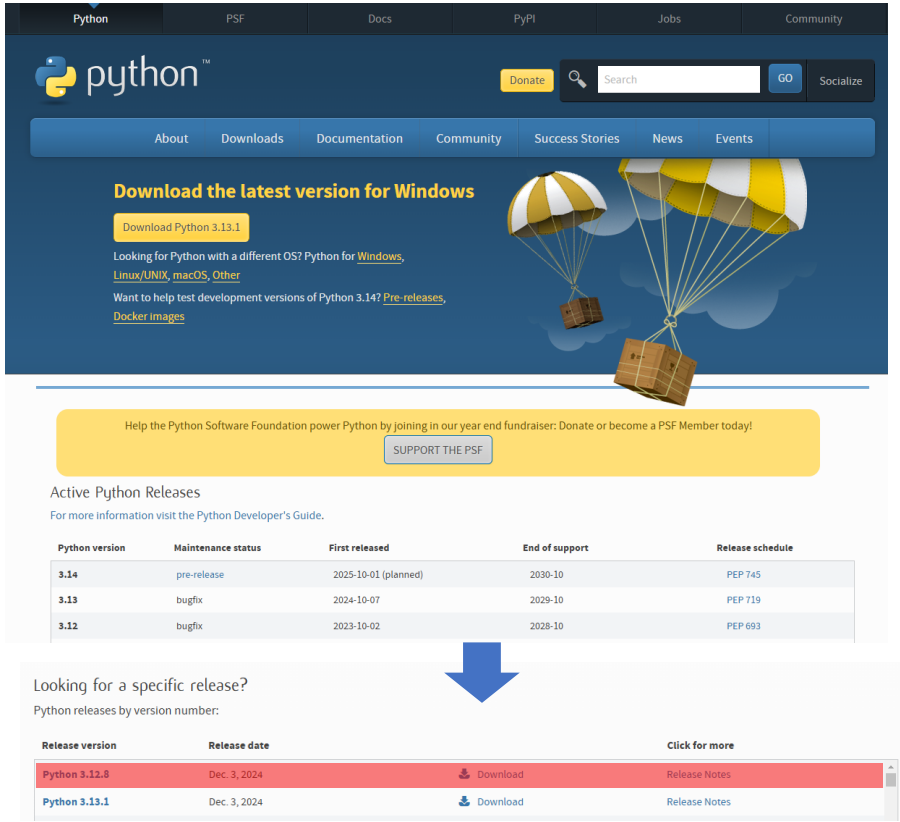
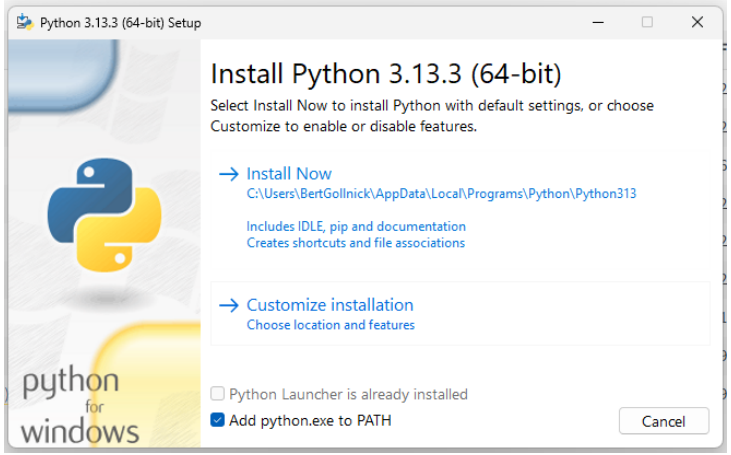
System Setup

Python Installation





Source: https://www.python.org/downloads/

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System Setup

Git – Option 1 (beginner)

GenerativeAIwithPython_Material Public

Watch 1

main | 1 Branch | 0 Tags | Go to file t | Add file | Code

Bert Gollnick agents updated

.vscode	model chats added
020_llm	README updated
040_vector_db	README updated
050_rag_baseline	README updated
060_rag_advanced	README updated
080_agents	agents updated

Local | Codespaces

Clone

HTTPS | SSH | GitHub CLI

https://github.com/DataScienceHamburg/Generati

Clone using the web URL.

Open with GitHub Desktop

Download ZIP

System Setup

Git – Option 2 (advanced)

1. Download Git and Install: https://git-scm.com/downloads

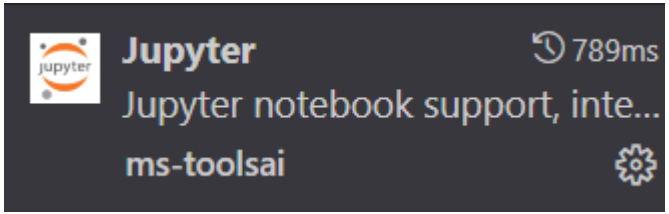
2. Configure git

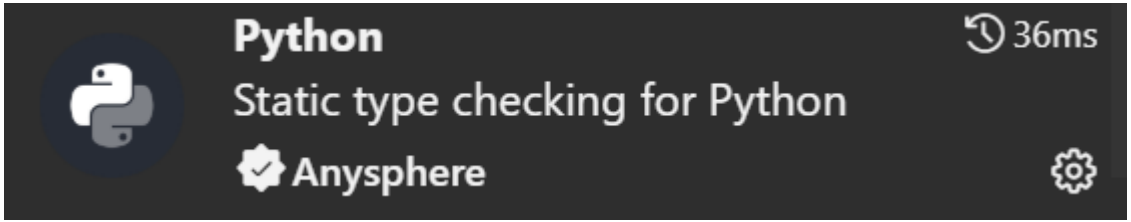
```
git config --global user.name "John Doe"

git config --global user.email johndoe@example.com
```

3. Clone Repo:
git clone https://github.com/GollnickDataSolutions/Bildungsurlaub_Python_ML_DS.git

System Setup

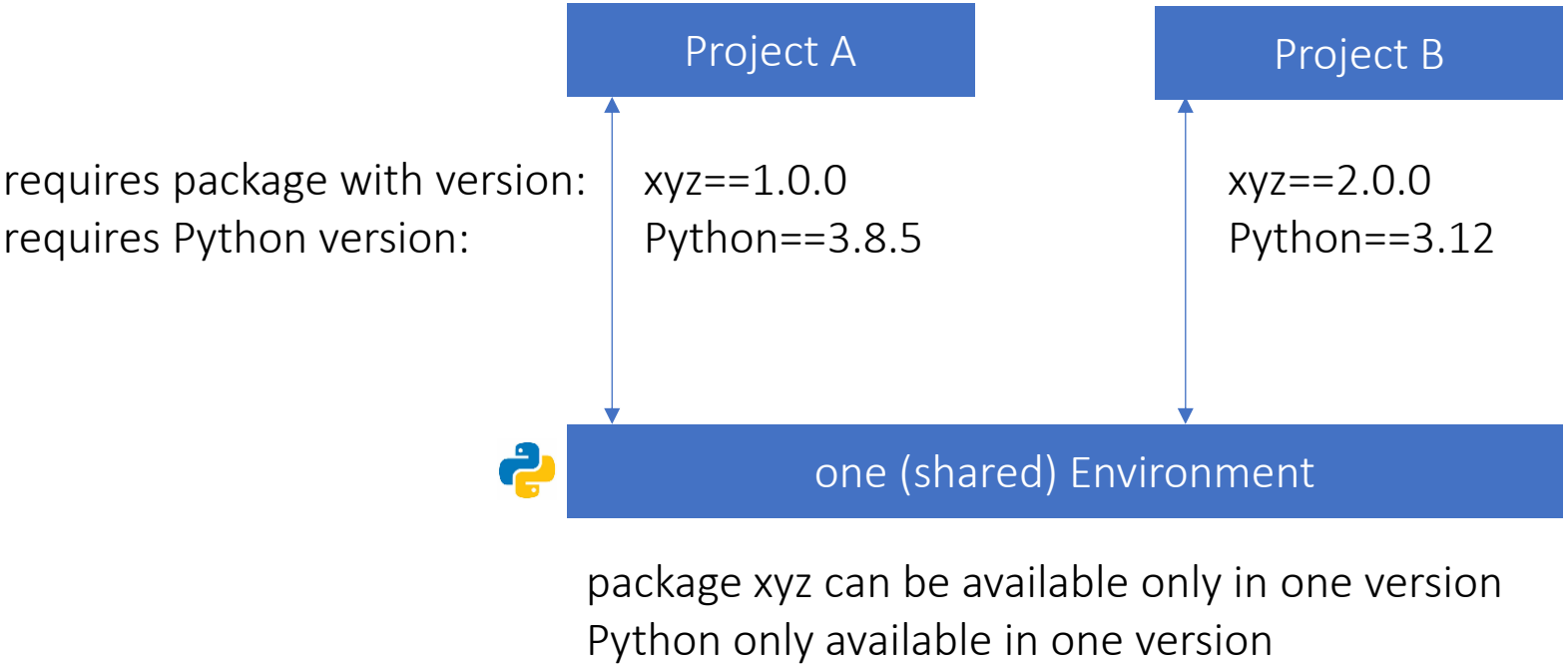
VS Code Extension





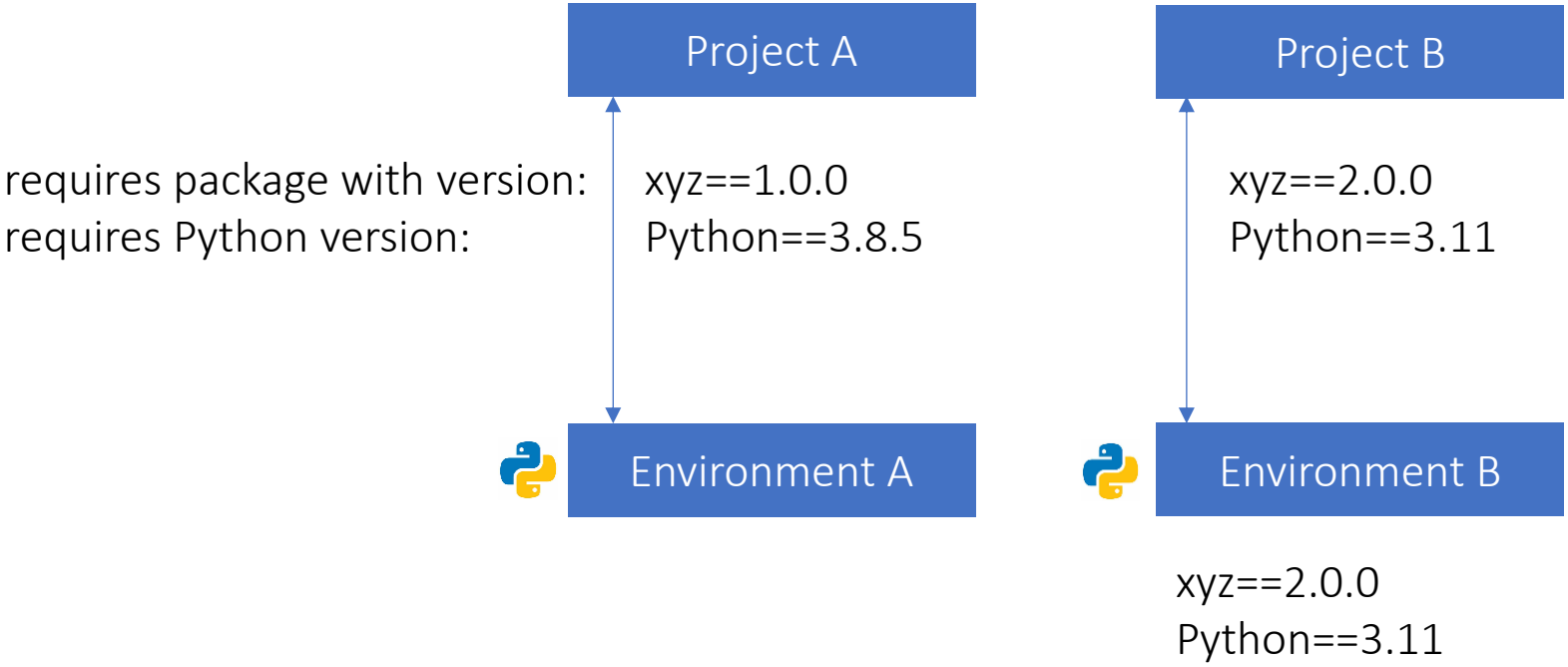
System Setup

Environment: Problem



System Setup

Environment: Solution



System Setup

Python Environment

An Environment is an isolated bundle of packages and Python versions.

It allows to

- avoid conflicts of packages
- easy setup for a project on another computer

- Available Environments:

Conda (requires Anaconda installation)	venv	Poetry	uv

System Setup

Python Environment setup with uv

uv

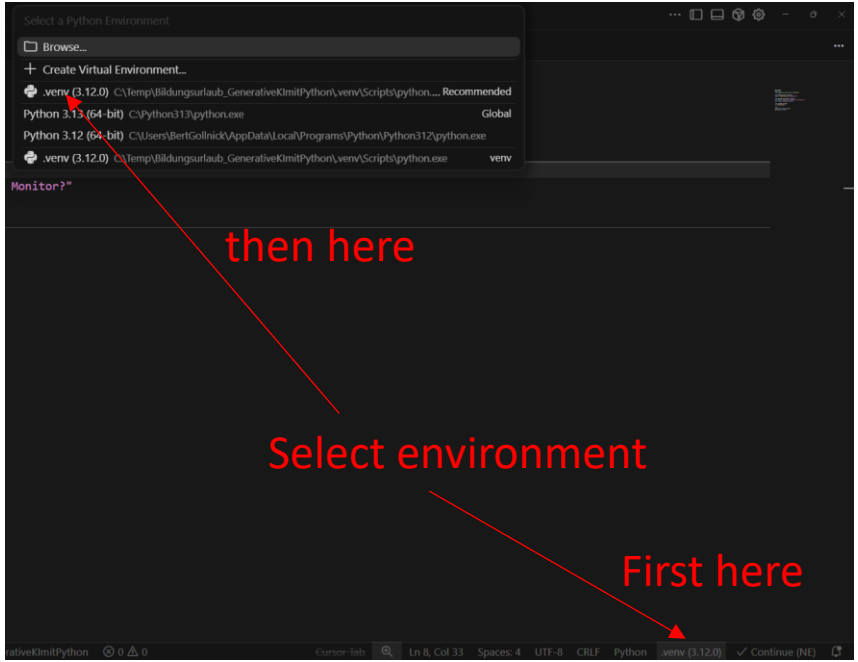
1. Install uv in general environment

   pip install uv

2. Setup of environment

   uv sync

3. Make sure that right environment is selected

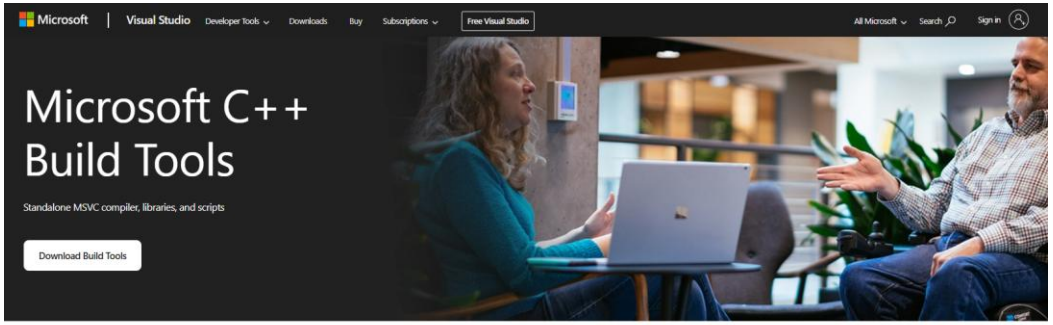


System Setup

Python Environment – venv

Windows: install C++ Build Tools

Mac: use faiss-cpu



The Microsoft C++ Build Tools provides MSVC toolsets via a scriptable, standalone installer without Visual Studio. Recommended if you build C++ libraries and applications targeting Windows from the command-line (e.g. as part of your continuous integration workflow). Includes tools shipped in Visual Studio 2015 Update 3, Visual Studio 2017, Visual Studio 2019, and latest version of Visual Studio 2022.

Walkthrough: Compiling a Native C++ Program on the Command Line →

How you can use the Build Tools

https://visualstudio.microsoft.com/visual-cpp-build-tools/

System Setup

OpenAI API Key

Personal / Course_VectorDbRagAgents

Playground Dashboa

SETTINGS
Your profile

ORGANIZATION
General
API keys
Admin keys
Members
Projects
Billing
Limits
Usage
Data controls
Verifications

PROJECT
General
API keys
Members
Limits

Billing

Overview Payment methods Billing history Preferences

Pay as you go

Credit balance
$7.75

Auto recharge is off
When your credit balance reaches $0, your API requests will stop working. Enable automatic recharge to automatically keep your balance topped up.
Enable auto recharge

Add to credit balance Cancel plan

Payment methods
Add or change payment method

Billing history
View past and current invoices

Preferences
Manage billing information

Usage limits
Set monthly spend limits

Pricing
View pricing and FAQs

Create new secret key

Owned by

You Service account

This API key is tied to your user and can make requests against the selected project. If you are removed from the organization or project, this key will be disabled.

Name Optional

My Test Key

Project

Select project

Permissions

All Restricted Read Only

Cancel Create secret key